

- Finding that our dataset is very imbalanced (<10% is sepsis positive)
 - Using XGboosted to help with the pos_weight
 - AND used SMOTE to synthetically create more sepsis positive cases by looking at distances in between two sepsis cases and choosing something in between
 - 129 to 1790 sepsis positive samples
- Results using their default values

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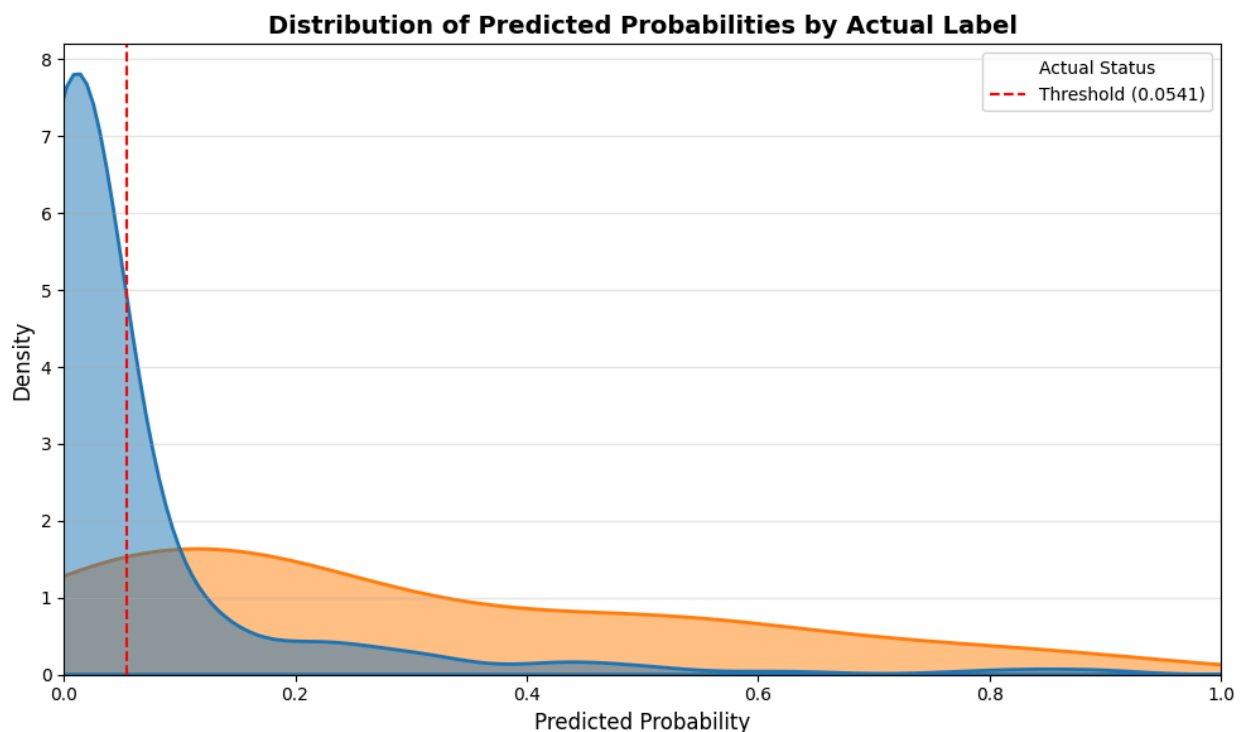
... --- Results (Decision Threshold: 0.0541) ---
      precision    recall  f1-score   support

   No Sepsis      0.98      0.78      0.87      431
     Sepsis       0.21      0.78      0.33       32

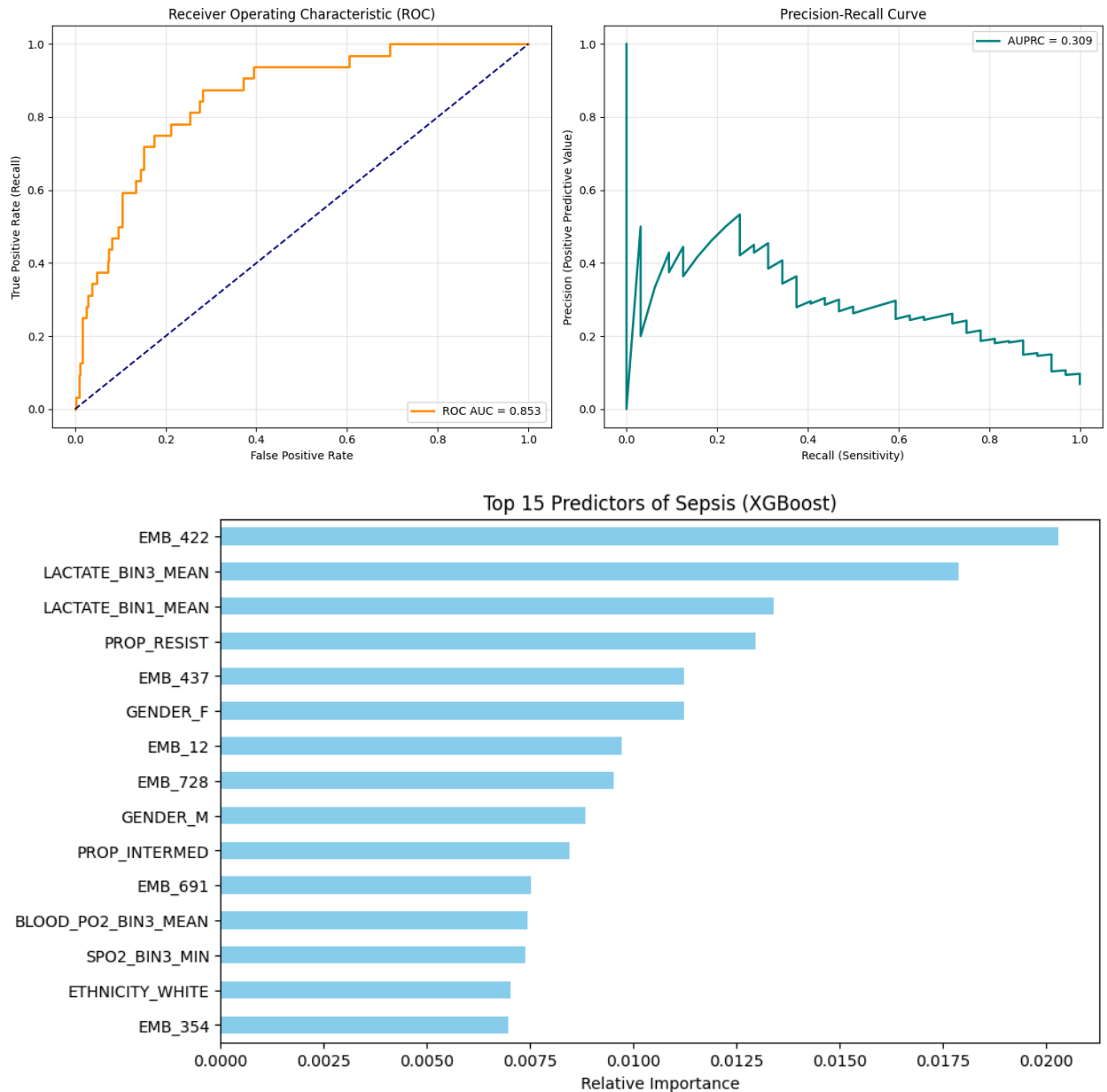
 accuracy          0.78      463
 macro avg       0.60      0.78      0.60      463
 weighted avg    0.93      0.78      0.83      463

F2-Score (Recall-focused): 0.5081

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* blue is no sepsis, orange is sepsis



- Took out NUM_ANTIBIOTICS column because it was the top predictor, but it probably occurs after a sepsis diagnosis IRL
- Shifted threshold from 0.5 (default) to 0.05 based on some algorithm to get 80% recall for sepsis positive
 - Meaning 80% of true positives is identified as true positive
- Random idea - if not for clinical use, this application can help create more balanced datasets by saving the positive flags (regardless of if it's actually positive, because its accuracy rate will still allow a more balanced dataset)